Course Plan

40-hour programme schedule

SpaceCDF -- AI-Assisted Concurrent Design Facility
University of Ottawa - SEDTI
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DRAFT

SpaceCDF Mission Design Course -- 40-Hour Programme

Course Title

Collaborative Space Mission Design: From Problem to Flight-Ready CubeSat

Audience

Engineers, scientists, and project managers participating in concurrent design facility (CDF) studies. No prior spacecraft design experience required but engineering background assumed.

Structure

- 40 contact hours over 5 days (8 hours/day) or 10 half-days
- Facilitator's Book: Complete reference with all content, solutions, diagrams
- Learner's Workbook: Condensed content with worksheets for tool exercises

Reference Documents

- NASA/SP-2016-6105 Rev 2 (Systems Engineering Handbook)
- NPR 7123.1D (NASA SE Processes)
- ECSS-E-ST-10C Rev.1 (Space Engineering General Requirements)
- ECSS-M-ST-10C Rev.1 (Project Management)
- ECSS-E-ST-10-02C Rev.1 (Verification)
- Wertz et al., Space Mission Engineering: The New SMAD (2011)
- Larson & Wertz, Space Mission Analysis and Design (SMAD4)
- Cal Poly CubeSat Design Specification Rev 14.1
- ITU Radio Regulations
- ECSS-U-AS-10C Rev.2 (Space Debris Mitigation)

Day 1: Mission Definition & Concept (8 hours)

Session 1.1: Introduction to Space Mission Design (2 hrs)

- What is a Concurrent Design Facility?

- The System-V model (NASA SEH Ch. 2-3)
- The 17 SE processes (NPR 7123.1)
- NASA/ECSS lifecycle phases and review gates
- Role of each CDF position
- Exercise: Map the 17 processes to the SpaceCDF tool

Session 1.2: Mission Need & Stakeholder Analysis (2 hrs)

- NASA SEH Process 1: Stakeholder Expectations Definition
- Problem statement writing (NASA SEH §4.1)
- Stakeholder identification and needs elicitation
- Objective definition with measurable success criteria
- MoE, MoP, TPM hierarchy
- Exercise: Define mission need for a sample problem (using SpaceCDF Step 1)
- Worksheet: Stakeholder matrix, objective hierarchy

Session 1.3: Mission Trade Analysis (2 hrs)

- NASA SEH Process 17: Decision Analysis
- Space vs non-space alternatives
- Trade study methodology: criteria, weightings, scoring
- Existing data services (Copernicus, Planet, Spire)
- When NOT to build a satellite
- Exercise: Run mission trade in SpaceCDF (Step 2)
- Worksheet: Trade study matrix with scoring

Session 1.4: Concept of Operations (2 hrs)

- NASA SEH Appendix S: ConOps structure
- Mission architecture: space, ground, user segments
- Mission phases (LEOP -> commissioning -> nominal -> disposal)
- Operational modes and duty cycling
- Data flow pipeline design
- Ground segment architecture options
- Exercise: Build ConOps in SpaceCDF
- Worksheet: ConOps outline per Appendix S

Day 2: Requirements & Functions (8 hours)

Session 2.1: Requirements Engineering (2 hrs)

- NASA SEH Process 2: Technical Requirements Definition
- SMART requirements (Specific, Measurable, Achievable, Relevant, Traceable)
- WHAT not HOW: requirements vs design choices
- Requirement hierarchy: mission -> system -> subsystem

- NASA SEH Appendix C: How to Write a Good Requirement
- Exercise: Generate requirements from objectives in SpaceCDF
- Worksheet: Requirements quality checklist (Appendix C)

Session 2.2: Functional Decomposition (2 hrs)

- NASA SEH Process 3: Logical Decomposition
- Function trees: objective -> function -> subfunction
- Allocation to subsystems (system boundary definition)
- Derived requirements from functions
- Performance criteria definition
- Exercise: Build function tree in SpaceCDF
- Worksheet: Function-to-requirement traceability matrix

Session 2.3: Interface Management (2 hrs)

- NASA SEH Process 12: Interface Management
- ECSS-E-ST-10-24C: Interface requirements
- N^2 interface matrix methodology
- Interface types: mechanical, electrical, thermal, data, RF, optical
- Conflict identification and resolution
- Exercise: Review interface matrix in SpaceCDF
- Worksheet: Interface specification for 2 subsystem pairs

Session 2.4: Design Budgets Introduction (2 hrs)

- Mass budget methodology (ECSS-E-HB-10-02A)
- Power budget methodology (ECSS-E-ST-20C)
- Margin philosophy by project phase
- Budget roll-up: component -> subsystem -> system
- Cost estimation approaches (parametric, analogy, bottom-up)
- Exercise: Review parametric budgets in SpaceCDF dashboard
- Worksheet: Mass budget template with margin policy

Day 3: Subsystem Design (8 hours)

Session 3.1: Orbit Design & Selection (2 hrs)

- Orbit mechanics fundamentals (Keplerian elements, perturbations)
- LEO/SSO/MEO/GEO/HEO trade-offs
- Coverage and revisit analysis
- Orbital lifetime and debris compliance (ECSS-U-AS-10C)
- Eclipse fraction and power implications
- Exercise: Run orbit trade in SpaceCDF
- Worksheet: Orbit selection trade matrix

Session 3.2: Payload & Communications (2 hrs)

- Optical payload sizing (GSD -> aperture -> mass -> power)
- RF payload sizing (link budget -> antenna -> power)
- SAR fundamentals (resolution -> antenna area -> power)
- Link budget methodology (ECSS-E-ST-50-05C)
- Frequency band selection and licensing
- Amateur vs experimental vs commercial licensing
- Exercise: Configure payload and review link budget
- Worksheet: Link budget calculation sheet

Session 3.3: Power, AOCS, Thermal (2 hrs)

- EPS architecture (solar array, battery, regulation)
- Power budget by operational mode with duty cycling
- Attitude control: magnetorquers vs reaction wheels vs star trackers
- Pointing budget (RSS error tree)
- Thermal control: passive vs active, radiator sizing
- Exercise: Review power/AOCS/thermal outputs in SpaceCDF
- Worksheet: Power mode budget table

Session 3.4: Structure, Propulsion, Data Handling (2 hrs)

- CubeSat Design Specification (CDS Rev 14.1)
- Structural design: launch loads, natural frequency, margin of safety
- Propulsion options: cold gas, electric, chemical
- Delta-V budget allocation
- OBC architecture, data storage, flight software
- PC/104 bus standard
- Exercise: Select equipment in SpaceCDF browser
- Worksheet: Component selection trade study

Day 4: Integration & Verification (8 hours)

Session 4.1: Equipment Selection & Integration (2 hrs)

- Component selection methodology (trade studies)
- RF chain compatibility (transponder + antenna band matching)
- Power bus compatibility (voltage, switched lines)
- Volume and mass fit verification
- Harness and cabling design
- Exercise: Complete equipment selection in SpaceCDF
- Worksheet: Equipment compatibility checklist

Session 4.2: Verification & Validation Planning (2 hrs)

- NASA SEH Processes 7-8: Verification & Validation
- ECSS-E-ST-10-02C: Verification methods (ATRI)
- Verification matrix structure
- Test levels: unit -> subsystem -> system -> acceptance -> qual
- Environmental testing (vibration, thermal-vac, EMC)
- Exercise: Review compliance matrix in SpaceCDF
- Worksheet: V&V matrix for 10 key requirements

Session 4.3: Risk Management (2 hrs)

- ECSS-M-ST-80C: Risk management process
- Risk identification, assessment (5x5 matrix), mitigation
- FMECA methodology (ECSS-Q-ST-30-02C)
- Single-point failure analysis
- Technical Performance Measures (TPMs)
- Exercise: Review risk scores and reliability in SpaceCDF
- Worksheet: Risk register for top 5 risks

Session 4.4: Cost & Schedule (2 hrs)

- Cost estimation: parametric (SSCM, COMPACT), analogy, bottom-up
- WBS structure (NPR 7120.5)
- CubeSat cost drivers and fractions
- Schedule estimation and critical path
- Learning curve effects for constellations
- Exercise: Review cost breakdown and run optimizer
- Worksheet: Cost estimate by WBS element

Day 5: Design Review & Regulatory (8 hours)

Session 5.1: Gate Review Preparation (2 hrs)

- ECSS-M-ST-10C: Review gate structure
- MCR/SRR/PDR/CDR exit criteria
- Design review presentation structure
- Action item management
- Exercise: Run gate review in SpaceCDF, resolve action items
- Worksheet: MCR presentation outline

Session 5.2: Regulatory & Licensing (2 hrs)

- Frequency licensing (ITU, IARU, national)
- ITU filing process (API -> coordination -> notification)
- Canadian RSSSA for remote sensing
- Export control (ITAR/EAR/CGP)

- UN Registration Convention (COPUOS)
- Space debris regulations (25yr/5yr rules)
- Exercise: Generate regulatory filings in SpaceCDF
- Worksheet: Licensing decision tree

Session 5.3: Launch Integration (2 hrs)

- Launch providers and pricing
- Deployer standards (ISIPOD, EXOpod, CSD)
- Launch ICD requirements (mechanical, electrical, environmental)
- Separation switches and inhibits
- Environmental test specification derivation
- Exercise: Select launch provider, review ICD requirements
- Worksheet: Launch ICD compliance checklist

Session 5.4: Design Optimisation & Final Review (2 hrs)

- Multi-objective optimisation (Pareto concepts)
- Sensitivity analysis (Morris screening)
- Design iteration and convergence
- Final design review presentation
- Lessons learned
- Exercise: Run optimizer, review Pareto front, select final design
- Final Exercise: Each team presents their complete mission design

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Assessment

- Day 1-4: Worksheets completed in class (formative)
- Day 5: Team presentation of complete mission design (summative)
- Post-course: Access to SpaceCDF tool and Facilitator's Book

Per-Position Deep Dives (Appendices)

Appendix A: Systems Engineer

- Budget management, margin policy, trade study leadership
- Cross-domain conflict resolution
- Gate review preparation and exit criteria evaluation

Appendix B: Mission Analyst

- Orbit mechanics, coverage analysis, constellation design
- Ground station selection, contact time analysis
- Lighting conditions, eclipse analysis

Appendix C: Payload Engineer

- Optical/RF/SAR payload sizing
- Payload data rates and storage requirements
- Instrument calibration and performance verification

Appendix D: Power Engineer

- Solar array sizing (body-mounted vs deployable)
- Battery sizing and cycle life
- EPS architecture and power distribution
- Power mode analysis with duty cycling

Appendix E: AOCS Engineer

- Attitude determination: sun sensors, star trackers, gyros
- Attitude control: magnetorquers, reaction wheels
- Pointing budget (RSS error analysis)
- Momentum management and dumping

Appendix F: Thermal Engineer

- Orbital thermal environment (hot case, cold case, eclipse)
- Passive thermal control (coatings, MLI, radiators)
- Active thermal control (heaters, heat pipes)
- Thermal margin policy (ECSS-E-ST-31C)

Appendix G: Communications Engineer

- Link budget fundamentals
- Frequency band selection and licensing
- Antenna types and sizing
- Modulation and coding selection
- Ground station network design

Appendix H: Propulsion Engineer

- Delta-V budget (orbit insertion, maintenance, deorbit, collision avoidance)
- Propulsion system types (cold gas, electric, chemical)
- Tank sizing and propellant management
- Thruster selection and performance

Appendix I: Structures Engineer

- CubeSat Design Specification compliance
- Launch load analysis (quasi-static, vibration, shock)
- Structural margin of safety calculation
- Mechanism design (deployment, separation)

Appendix J: Cost Engineer

- Parametric cost models (SSCM, COMPACT, NICM)
- WBS development and cost allocation
- Risk-adjusted cost (Monte Carlo)
- Learning curve for production runs

Appendix K: Compliance / Regulatory Engineer

- ECSS standard applicability and tailoring
- Frequency licensing decision tree
- Export control classification
- Space debris compliance assessment
- End-of-life planning and filing

Appendix L: Ground Segment Engineer

- Ground station architecture
- Mission control centre design
- Data processing pipeline
- Operations concept and staffing

Appendix M: Software Engineer

- Flight software architecture
- FDIR (Fault Detection, Isolation, Recovery)
- Telecommand/telemetry definition
- Ground software and operations tools

Document Production Plan

Facilitator's Book (~400 pages)

- Full content for all 20 sessions
- Complete solutions to all exercises
- Diagrams and formulae throughout
- Reference citations per section
- Comprehensive index

Learner's Workbook (~150 pages)

- Condensed content per session (key concepts + references)
- 20 worksheets (one per session)
- Tool exercise guides (step-by-step SpaceCDF instructions)
- Space for notes and discussion points

- Appendix with key formulae and reference tables